Bézier Curve — Matrix Method (Straight)

$$P(u) = U \cdot M \cdot G$$

Explanation of the Program Output

Case 2: Quartic 3-D, 5 Control Points

`04-bezier-matrix-straight.jl`

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This document explains the output of `04-bezier-matrix-straight.jl` when run with `CASE = 2`. The mathematical method is identical to script 03 ($P(u) = U \cdot M \cdot G$, same control points, same matrices, same results). What changed is the **output format**: script 03 was pedagogical — it derived M column by column, showed each dot product explicitly, and printed five numbered steps. Script 04 is straight — it prints the final matrices and result directly, with no intermediate working shown.

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1. What Changed from Script 03

Same curve. Same matrices. Same numbers. Just a cleaner, more direct presentation.

Script 03 had five numbered steps, each showing the derivation or computation in detail. Script 04 has five *sections* — flat, unnumbered, one per matrix or result — printed without any working.

Script 03 output block	Script 04 output block	Content
Step 1: Matrix Formulation	*(header only)*	Formula and shapes
Step 2: Derive M	M — Basis Matrix	Just the matrix
Step 3: Compute $M \cdot G$	$M \cdot G$ — Polynomial Coefficient Matrix	Just the matrix
(inside Step 3)	Resulting Polynomials	The polynomial strings
Step 4: Evaluate $U \cdot (M \cdot G)$	*(dropped)*	Per-u dot products
Step 5: Summary table	Evaluation Table	Box-bordered table
(no equivalent)	G — Geometry Matrix	Control points as matrix

One new block: G is now shown explicitly. Script 03 listed the control points as a plain list in Step 1. Script 04 prints them as the labelled geometry matrix G — the same data, formatted as a proper matrix table, making the connection to $P(u) = U \cdot M \cdot G$ more direct.

One dropped block: the per-u dot product display. Script 03 Step 4 showed every multiplication $U[k] \cdot (MG)[k]$ for each u value. Script 04 drops this entirely and goes straight to the summary table. The numbers are the same; you just no longer see how they were reached.

2. Header

```
====================================================================
  Bezier Curve - Matrix Method  [P(u) = U * M * G]
  Case 2  |  Quartic  |  3D  |  5 Control Points
====================================================================
```

Identical to script 03. The formula $P(u) = U \cdot M \cdot G$ is stated in the title line, setting the context for the four matrices that follow.

3. G — Geometry Matrix

```
  G  - Geometry Matrix (Control Points)
  --------------------------------------------------------------


      x  y  z
      -  -  -
```

```
P0  0  0  1
P1  0  4  1
P2  4  0  1
P3  4  4  1
P4  5  4  1
```

G is the 5×3 control point matrix:

$$G = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 4 & 1 \\ 4 & 0 & 1 \\ 4 & 4 & 1 \\ 5 & 4 & 1 \end{bmatrix}$$

Row i = control point P_{i-1}. Column j = axis (x, y, z). This is the same data as the control point list in the header of scripts 01–03, now presented as the matrix it actually is in the computation $P(u) = U \cdot M \cdot G$.

All z-coordinates are 1, which means the curve lies entirely in the plane $z = 1$. This will be confirmed when $z(u) = 1$ appears in every result below.

4. M — Basis Matrix

```
M  - Basis Matrix (derived from Bernstein)
-----------------------------------------------------------------

     P0   P1   P2  P3  P4
     --  ---  ---  --  --
u^4   1   -4    6  -4   1
u^3  -4   12  -12   4   0
u^2   6  -12    6   0   0
u^1  -4    4    0   0   0
u^0   1    0    0   0   0
```

M is the 5×5 basis matrix derived from the Bernstein polynomials. It is the same matrix that script 03 derived column by column in Step 2. Here it is printed directly without derivation.

$$M = \begin{bmatrix} 1 & -4 & 6 & -4 & 1 \\ -4 & 12 & -12 & 4 & 0 \\ 6 & -12 & 6 & 0 & 0 \\ -4 & 4 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

The column labels `P0` through `P4` indicate that column i holds the monomial coefficients of $B(4, i, u)$, the i-th Bernstein basis polynomial. The row labels indicate the power of u each row represents, from u^4 (top) to $u^0 = 1$ (bottom).

Reading M: Column 0 contains $[1, -4, 6, -4, 1]$ — the coefficients of $(1-u)^4 = 1 - 4u + 6u^2 - 4u^3 + u^4$, reading from the u^4 row downward. Column 4 contains $[1, 0, 0, 0, 0]$ — the coefficient of u^4 only, matching $B(4, 4, u) = u^4$. The lower-left triangle is all zeros because $B(4, i, u)$ contains u^i as its lowest power.

5. $M \cdot G$ — Polynomial Coefficient Matrix

```
M * G  - Polynomial Coefficient Matrix
-----------------------------------------------------------------


       x    y  z
     ---  ---  -
u^4   13  -28  0
u^3  -32   64  0
u^2   24  -48  0
u^1    0   16  0
u^0    0    0  1
```

$M \cdot G$ is the 5×3 product of the basis matrix and the geometry matrix. Row k gives the coefficient of u^{4-k} for each axis.

$$M \cdot G = \begin{bmatrix} 13 & -28 & 0 \\ -32 & 64 & 0 \\ 24 & -48 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Reading each column:

- x **column:** $[13, -32, 24, 0, 0]^T$ — coefficients of $13u^4 - 32u^3 + 24u^2$.
- y **column:** $[-28, 64, -48, 16, 0]^T$ — coefficients of $-28u^4 + 64u^3 - 48u^2 + 16u$.
- z **column:** $[0, 0, 0, 0, 1]^T$ — constant polynomial $z(u) = 1$.

Verify the x column by hand:

$M \cdot G_x = M \cdot [0, 0, 4, 4, 5]^T$. Multiplying row by row:

Row 0 (u^4): $1(0) + (-4)(0) + 6(4) + (-4)(4) + 1(5) = 0 + 0 + 24 - 16 + 5 = 13$✓

Row 1 (u^3): $(-4)(0) + 12(0) + (-12)(4) + 4(4) + 0(5) = 0 + 0 - 48 + 16 + 0 = -32$✓

Row 2 (u^2): $6(0) + (-12)(0) + 6(4) + 0(4) + 0(5) = 0 + 0 + 24 + 0 + 0 = 24$✓

6. Resulting Polynomials

```
Resulting Polynomials
-----------------------------------------------------------------

x(u)  =  13u^4  - 32u^3  + 24u^2
y(u)  =  -28u^4  + 64u^3  - 48u^2  + 16u
z(u)  =  1
```

These are the final monomial polynomials, read directly from the columns of $M \cdot G$:

$$x(u) = 13u^4 - 32u^3 + 24u^2$$
$$y(u) = -28u^4 + 64u^3 - 48u^2 + 16u$$
$$z(u) = 1$$

These are identical to the polynomials produced by all previous scripts. The three methods (algebraic expansion, symbolic CAS, matrix method) converge to the same result.

Quick sanity checks:

$x(0) = 0$, $x(1) = 13 - 32 + 24 = 5$. The curve starts at $x = 0$ and ends at $x = 5 = P_4^x$. ✓

$y(0) = 0$, $y(1) = -28 + 64 - 48 + 16 = 4$. The curve starts at $y = 0$ and ends at $y = 4 = P_4^y$. ✓

$z(u) = 1$ for all u. The curve is entirely in the plane $z = 1$. ✓

7. Evaluation Table

```
Evaluation Table  -  P(u) = U * (M * G)
-----------------------------------------------------------------

+=============================================================+
| u       x(u)                y(u)                z(u)        |
|=============================================================|
| 0       0  (0.0000)         0  (0.0000)    1  (1.0000)      |
| 0.2500  269//256  (1.0508)  121//64 (1.8906)  1  (1.0000)   |
| 0.5000  45//16  (2.8125)    9//4  (2.2500)    1  (1.0000)   |
| 0.7500  1053//256 (4.1133)  201//64 (3.1406)  1  (1.0000)   |
| 1       5  (5.0000)         4  (4.0000)        1  (1.0000)  |
+=============================================================+
```

The evaluation table is identical in content to script 03's Step 5 summary table. Each cell shows the exact rational result followed by the 4-decimal-place float in parentheses.

u	$x(u)$ exact	$x(u)$ decimal	$y(u)$ exact	$y(u)$ decimal
0	0	0.0000	0	0.0000
0.25	269/256	1.0508	121/64	1.8906
0.50	45/16	2.8125	9/4	2.2500
0.75	1053/256	4.1133	201/64	3.1406
1	5	5.0000	4	4.0000

Verify $x(0.5) = 45/16$:

At $u = 0.5$, the U vector is $[\frac{1}{16}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2}, 1]$. The x column of $M \cdot G$ is $[13, -32, 24, 0, 0]$.

$$\tfrac{1}{16}(13) + \tfrac{1}{8}(-32) + \tfrac{1}{4}(24) + \tfrac{1}{2}(0) + 1(0) = \tfrac{13}{16} - 4 + 6 = \tfrac{13}{16} + 2 = \tfrac{45}{16}$$

$45 \div 16 = 2.8125$. ✓

8. Summary: Script 03 vs Script 04

The same computation, two different reading experiences.

Aspect	Script 03	Script 04
Purpose	Teach the method step by step	Show the result concisely
M derivation	Shown column by column	Not shown
Dot products	Shown per u value	Not shown
G display	As a plain list	As a labelled matrix
Output length	Long — every step visible	Short — results only
Best used for	First reading, learning	Reference, verification

Which should you use?

Use script 03 when you want to understand *why* the numbers come out the way they do — when you are learning the matrix method or debugging.

Use script 04 when you already understand the method and just want to *see* the matrices and the final curve polynomials quickly.

Script 03 is the lecture. Script 04 is the cheat sheet. Keep both.